

Run: rk4-frz-a0c0e-300e

Log-log loss curves

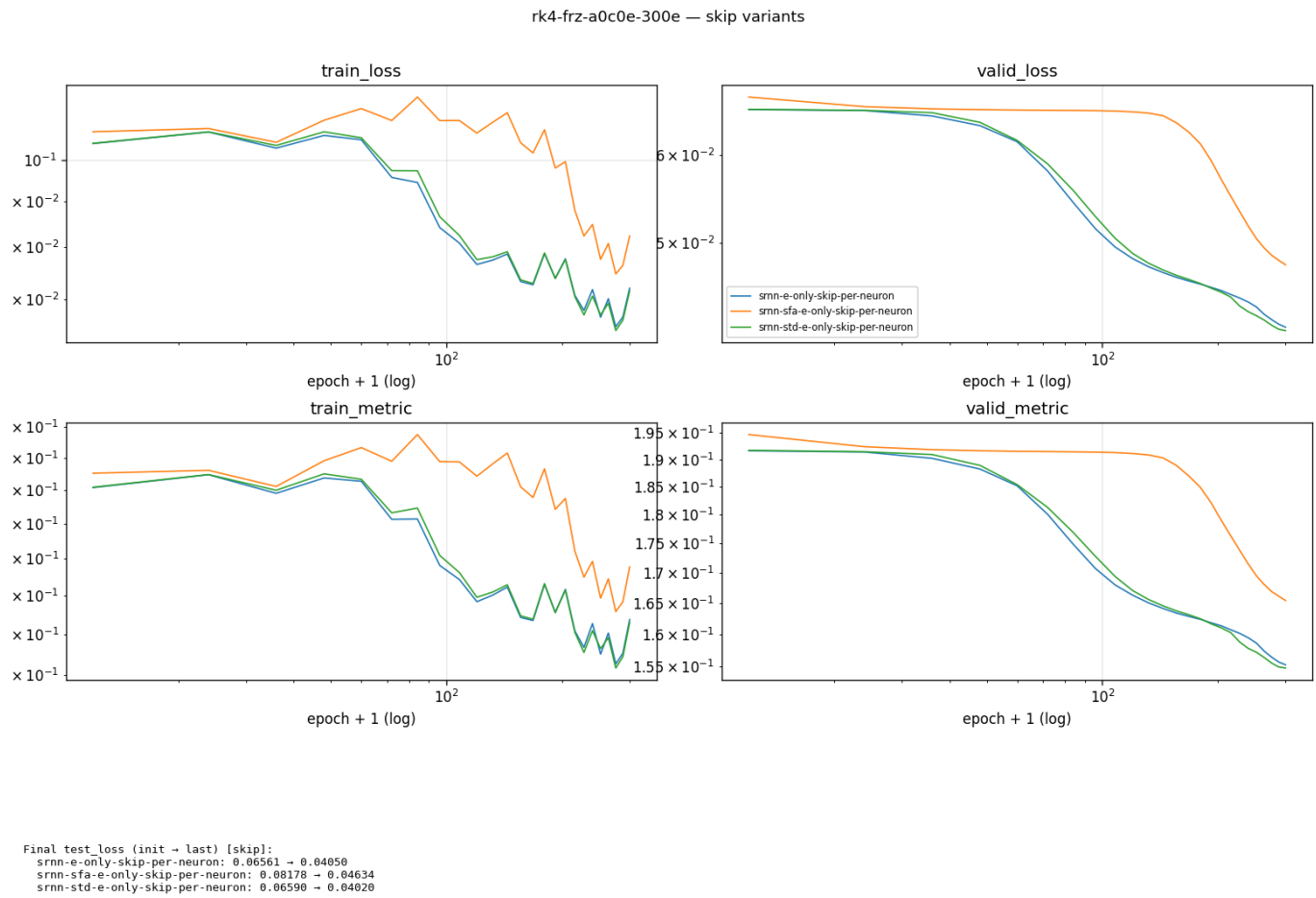
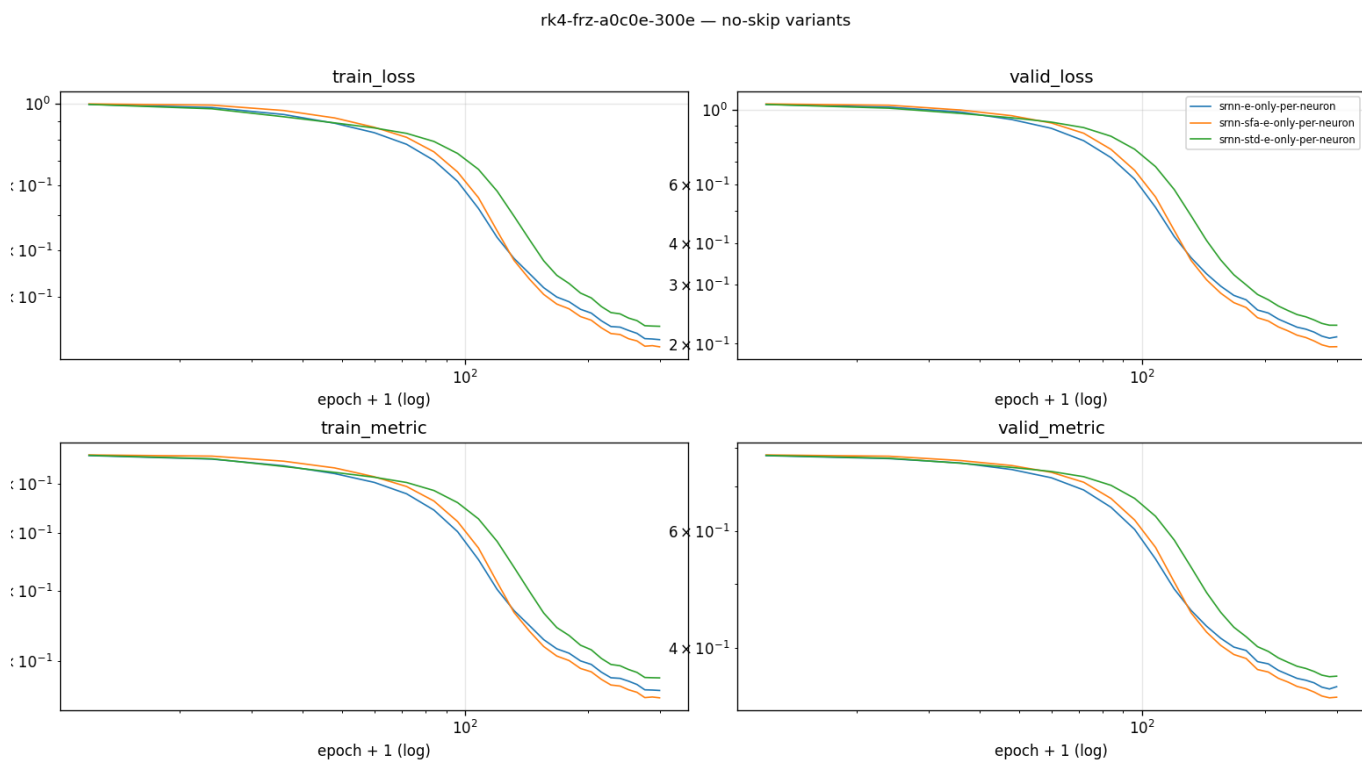


Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.90042 - 0.19152
 srnn-sfa-e-only-per-neuron: 0.91618 - 0.18050
 srnn-std-e-only-per-neuron: 0.90077 - 0.20773

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

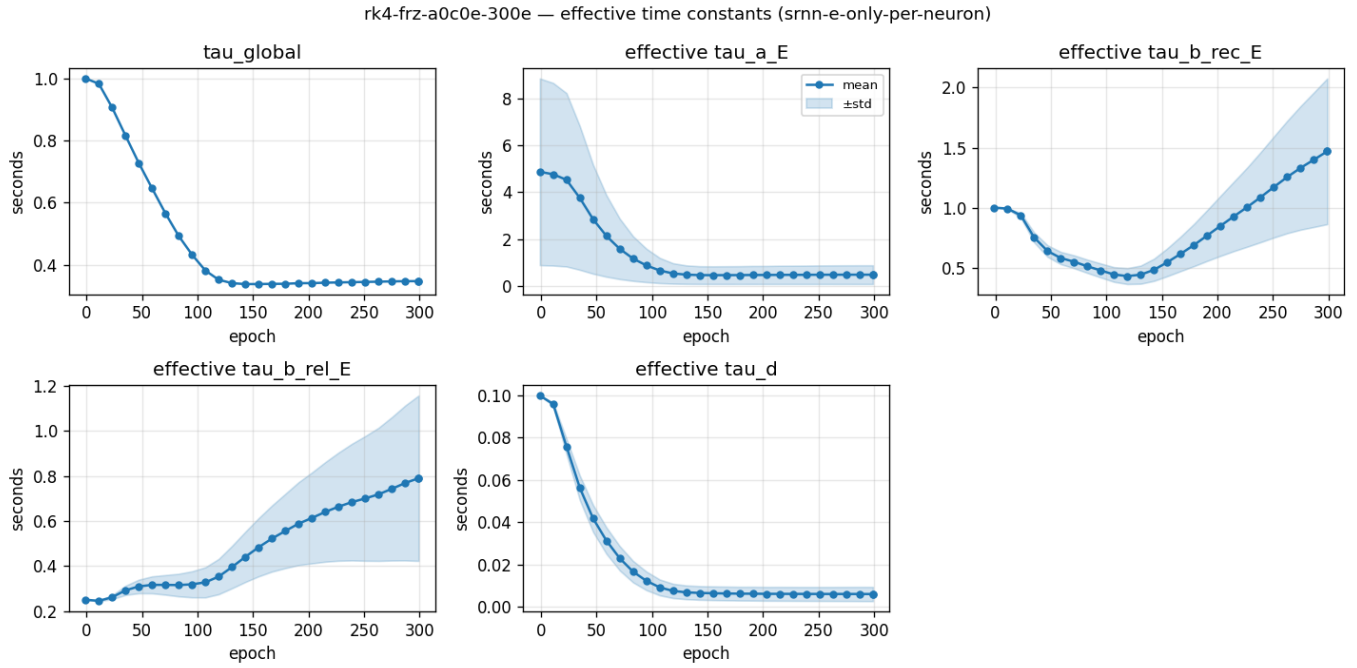
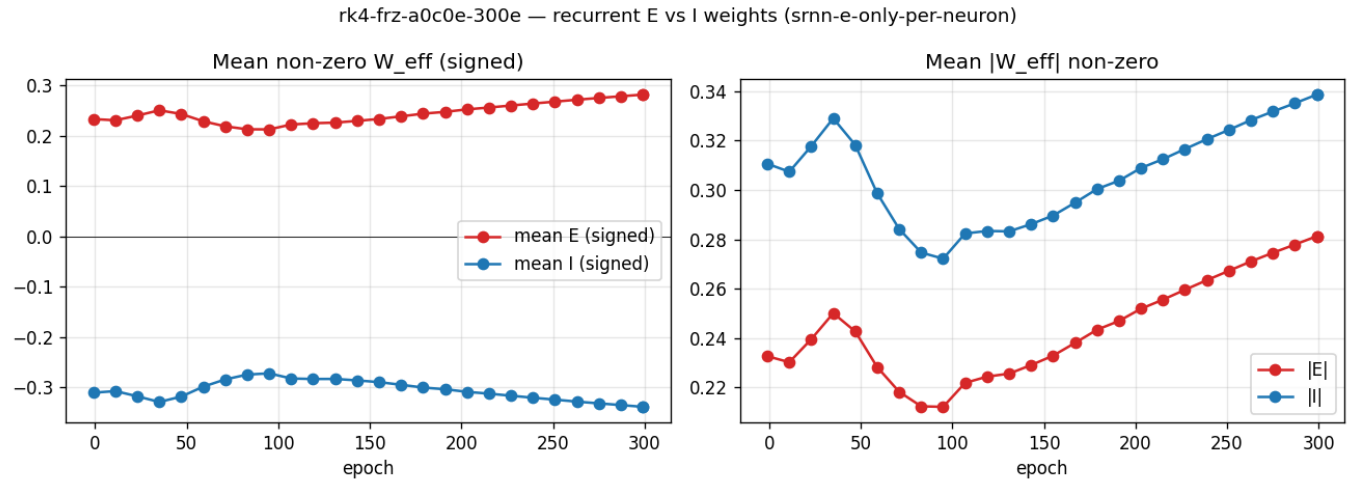


Figure 3: Tau evolution

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# rk4-frz-a0c0e-300e - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.34698	-	-	scalar
tau_d (s)	+0.1	+0.005996	0	0.003382	300
tau_a_E[j=0] (s)	+0.25	+0.012352	0	0.006063	150
tau_a_E[j=1] (s)	+4.3832	+0.4238	0	0.07366	150
tau_a_E[j=2] (s)	+10	+0.98354	0	0.0621	150
tau_b_rec_E (s)	+1	+1.4691	0	0.6074	150
tau_b_rel_E (s)	+0.25	+0.78938	0	0.3671	150
W_eff (E src, signed)	+0.23267	+0.28128	0.07641	0.1339	15048
W_eff (I src, signed)	-0.31062	-0.33865	0.07724	0.1118	14977
W_eff (E src)	+0.23267	+0.28128	0.07641	0.1339	15048
W_eff (I src)	+0.31062	+0.33865	0.07724	0.1118	14977
W_in_eff (input neurons)	-0.0003386	-5.5226e-05	0.1012	0.03547	6675
c_E[j=0] (SFA coupling)	+0.05	+0.12817	0	0.09836	150
c_E[j=1] (SFA coupling)	+0.05	+0.03279	0	0.0176	150
c_E[j=2] (SFA coupling)	+0.05	+0.034775	0	0.01713	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	-2.8854e-05	+0.0019011	0.06604	0.3711	6675
readout_bias	+0	-0.0090848	0	0.1653	89
ic.ic (per-variant init)	+0.10233	+0.10233	0.3832	0.3832	900

srnn-e-only-skip-per-neuron

rk4-frz-a0c0e-300e — effective time constants (srnn-e-only-skip-per-neuron)

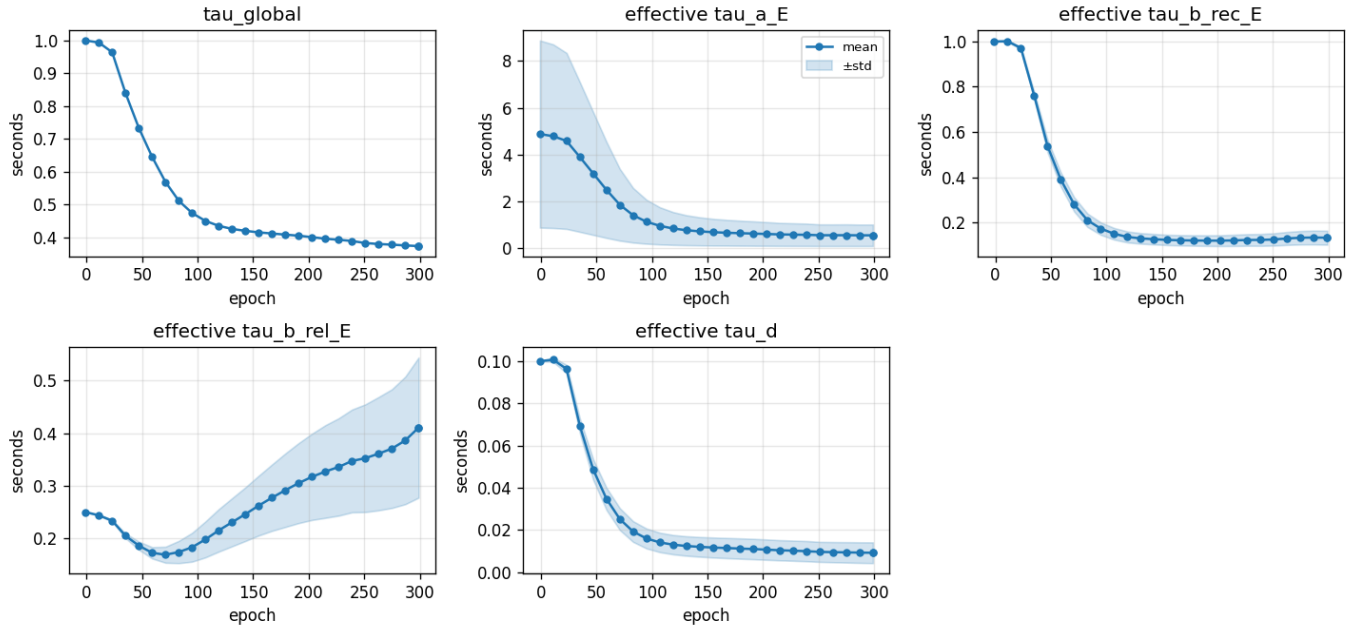
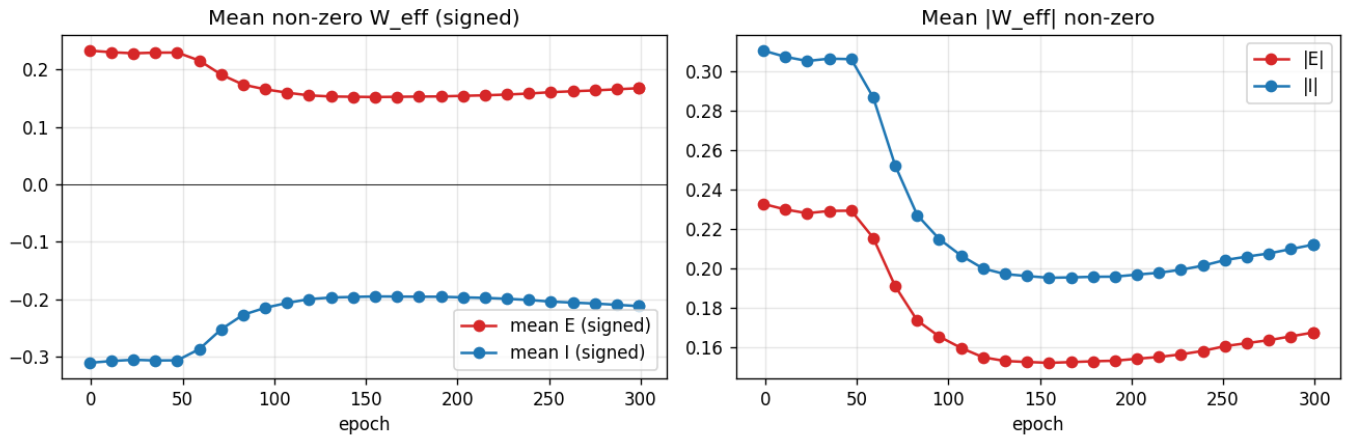


Figure 5: Tau evolution

rk4-frz-a0c0e-300e — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

```
# rk4-frz-a0c0e-300e - variant srnn-e-only-skip-per-neuron (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.37391	-	-	scalar
tau_d (s)	+0.1	+0.0092588	0	0.00493	300
tau_a_E[j=0] (s)	+0.25	+0.017057	0	0.008919	150
tau_a_E[j=1] (s)	+4.3832	+0.50498	0	0.04034	150
tau_a_E[j=2] (s)	+10	+1.1382	0	0.02469	150
tau_b_rec_E (s)	+1	+0.13224	0	0.03099	150
tau_b_rel_E (s)	+0.25	+0.41067	0	0.1333	150
W_eff (E src, signed)	+0.23267	+0.16725	0.07641	0.06634	15048
W_eff (I src, signed)	-0.31062	-0.21189	0.07724	0.06713	14977
W_eff (E src)	+0.23267	+0.16725	0.07641	0.06634	15048
W_eff (I src)	+0.31062	+0.21189	0.07724	0.06713	14977
W_in_eff (input neurons)	-0.00084237	-0.00022745	0.1008	0.01649	6675
c_E[j=0] (SFA coupling)	+0.05	+0.061055	0	0.03309	150
c_E[j=1] (SFA coupling)	+0.05	+0.032305	0	0.009822	150
c_E[j=2] (SFA coupling)	+0.05	+0.033062	0	0.00887	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00078219	-0.0088944	0.06654	0.1712	6675
readout_bias	+0	+0.096091	0	0.127	89
ic.ic (per-variant init)	+0.10092	+0.10092	0.3832	0.3832	900

srnn-sfa-e-only-per-neuron

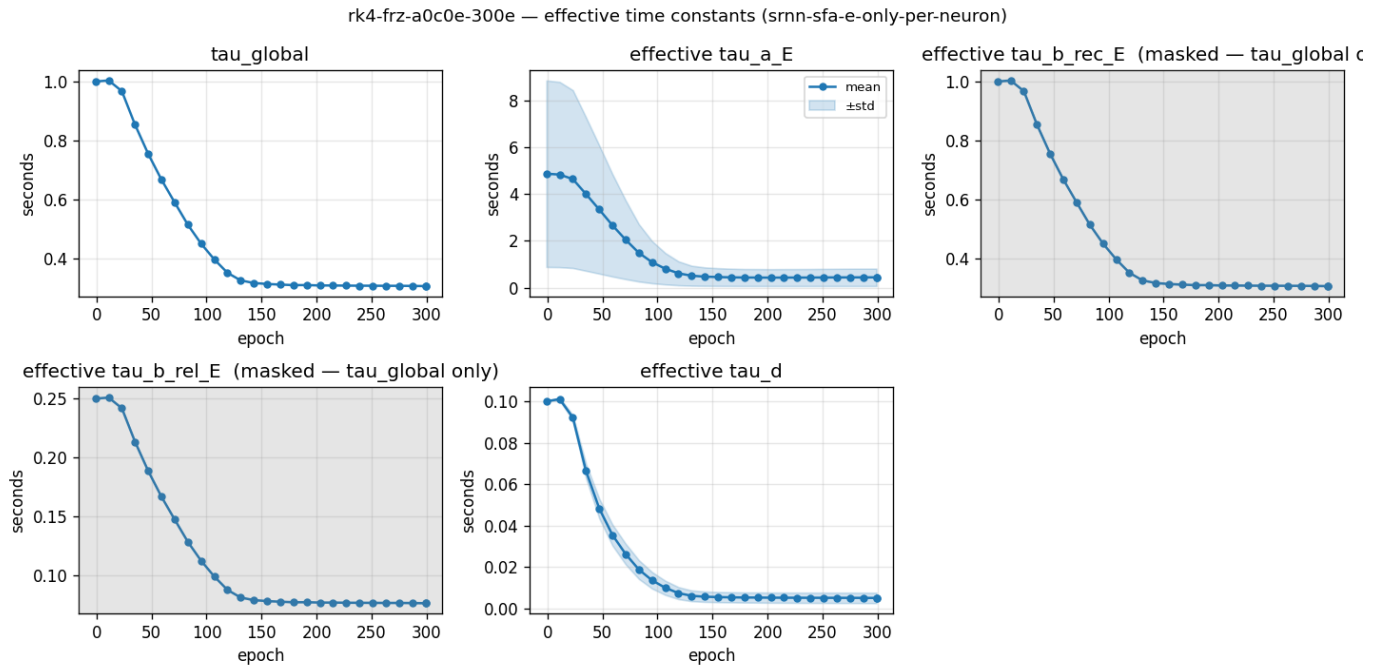
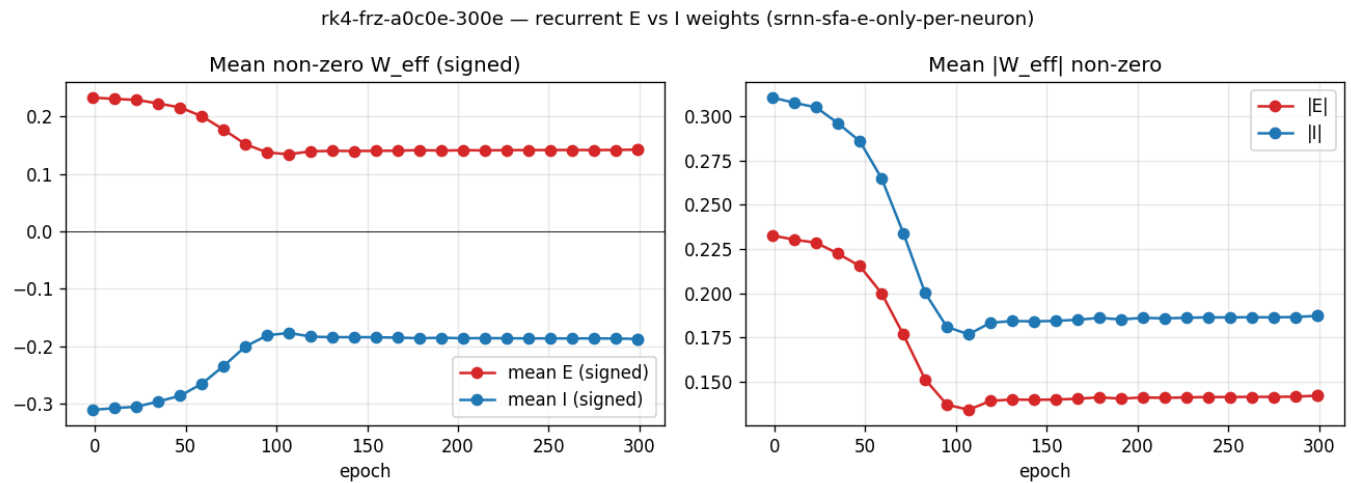


Figure 7: Tau evolution

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

rk4-frz-a0c0e-300e - variant srnn-sfa-e-only-per-neuron (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.30672	-	-	scalar
tau_d (s)	+0.1	+0.0050946	0	0.002524	300
tau_a_E[j=0] (s)	+0.25	+0.012626	0	0.005371	150
tau_a_E[j=1] (s)	+4.3832	+0.389	0	0.06346	150
tau_a_E[j=2] (s)	+10	+0.91717	0	0.05339	150
tau_b_rec_E (s)	+1	+0.30672	0	0	150
tau_b_rel_E (s)	+0.25	+0.076681	0	0	150
W_eff (E src, signed)	+0.23267	+0.14209	0.07641	0.05529	15048
W_eff (I src, signed)	-0.31062	-0.1872	0.07724	0.05725	14977
W_eff (E src)	+0.23267	+0.14209	0.07641	0.05529	15048
W_eff (I src)	+0.31062	+0.1872	0.07724	0.05725	14977
W_in_eff (input neurons)	-0.00015382	-0.00060358	0.09934	0.03697	6675
c_E[j=0] (SFA coupling)	+0.05	+0.16302	0	0.08614	150
c_E[j=1] (SFA coupling)	+0.05	+0.078942	0	0.02105	150
c_E[j=2] (SFA coupling)	+0.05	+0.077687	0	0.01942	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	-0.00098972	+0.0033037	0.06678	0.2511	6675
readout_bias	+0	-0.056828	0	0.1019	89
ic.ic (per-variant init)	+0.098719	+0.098719	0.4415	0.4415	900

srnn-sfa-e-only-skip-per-neuron

rk4-frz-a0c0e-300e — effective time constants (srnn-sfa-e-only-skip-per-neuron)

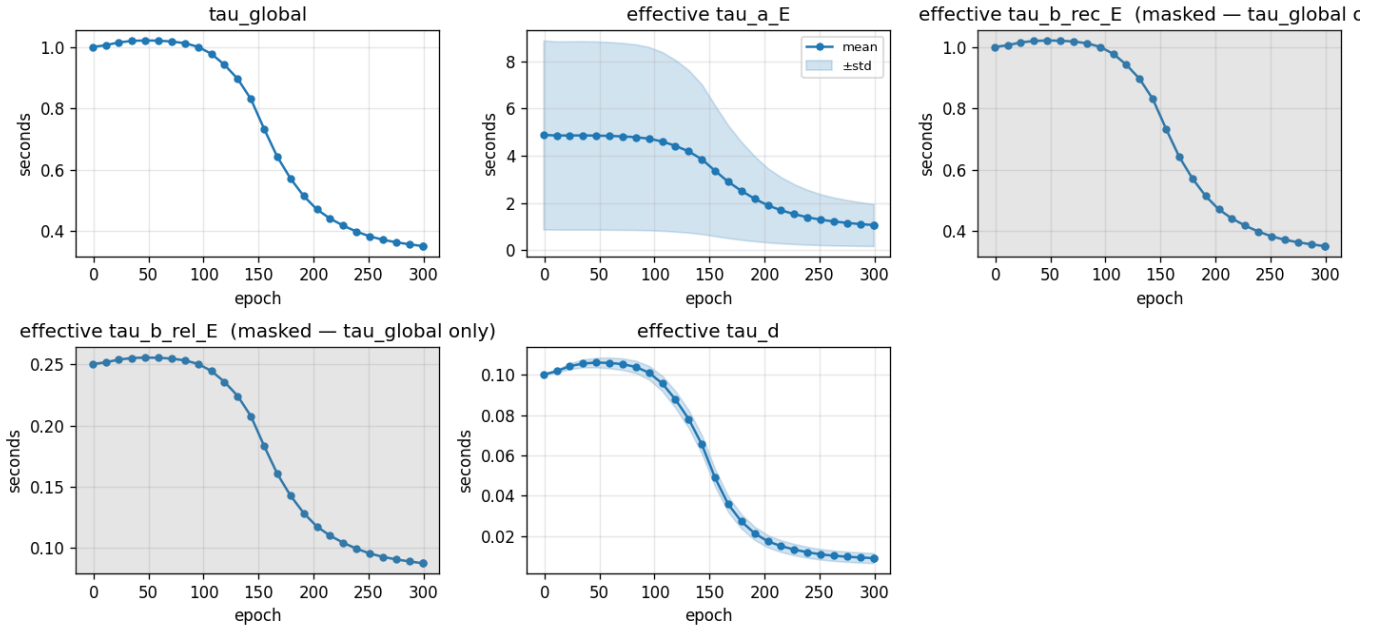
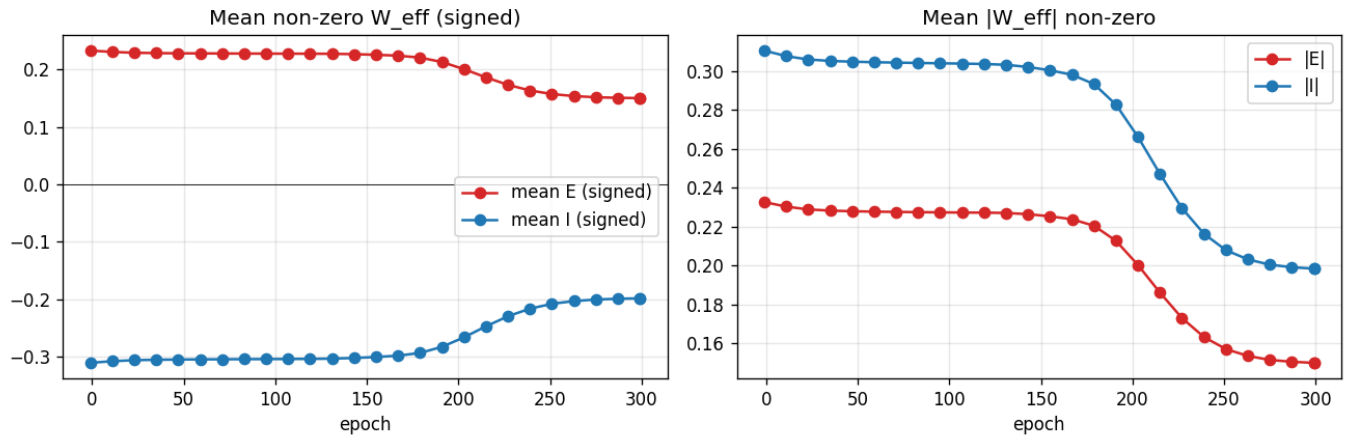


Figure 9: Tau evolution

rk4-frz-a0c0e-300e — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# rk4-frz-a0c0e-300e - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.35111	-	-	scalar
tau_d (s)	+0.1	+0.0091446	0	0.002498	300
tau_a_E[j=0] (s)	+0.25	+0.045267	0	0.01344	150
tau_a_E[j=1] (s)	+4.3832	+0.94338	0	0.05165	150
tau_a_E[j=2] (s)	+10	+2.211	0	0.02075	150
tau_b_rec_E (s)	+1	+0.35111	0	0	150
tau_b_rel_E (s)	+0.25	+0.087779	0	0	150
W_eff (E src, signed)	+0.23267	+0.14995	0.07641	0.05247	15048
W_eff (I src, signed)	-0.31062	-0.19841	0.07724	0.05476	14977
W_eff (E src)	+0.23267	+0.14995	0.07641	0.05247	15048
W_eff (I src)	+0.31062	+0.19841	0.07724	0.05476	14977
W_in_eff (input neurons)	-0.0011621	-0.00028593	0.09962	0.03203	6675
c_E[j=0] (SFA coupling)	+0.05	+0.052788	0	0.01142	150
c_E[j=1] (SFA coupling)	+0.05	+0.043313	0	0.008855	150
c_E[j=2] (SFA coupling)	+0.05	+0.043164	0	0.009024	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00016801	+0.00093434	0.0673	0.1085	6675
readout_bias	+0	-0.0040811	0	0.07401	89
ic.ic (per-variant init)	+0.12557	+0.12557	0.42	0.42	900

srnn-std-e-only-per-neuron

rk4-frz-a0c0e-300e — effective time constants (srnn-std-e-only-per-neuron)

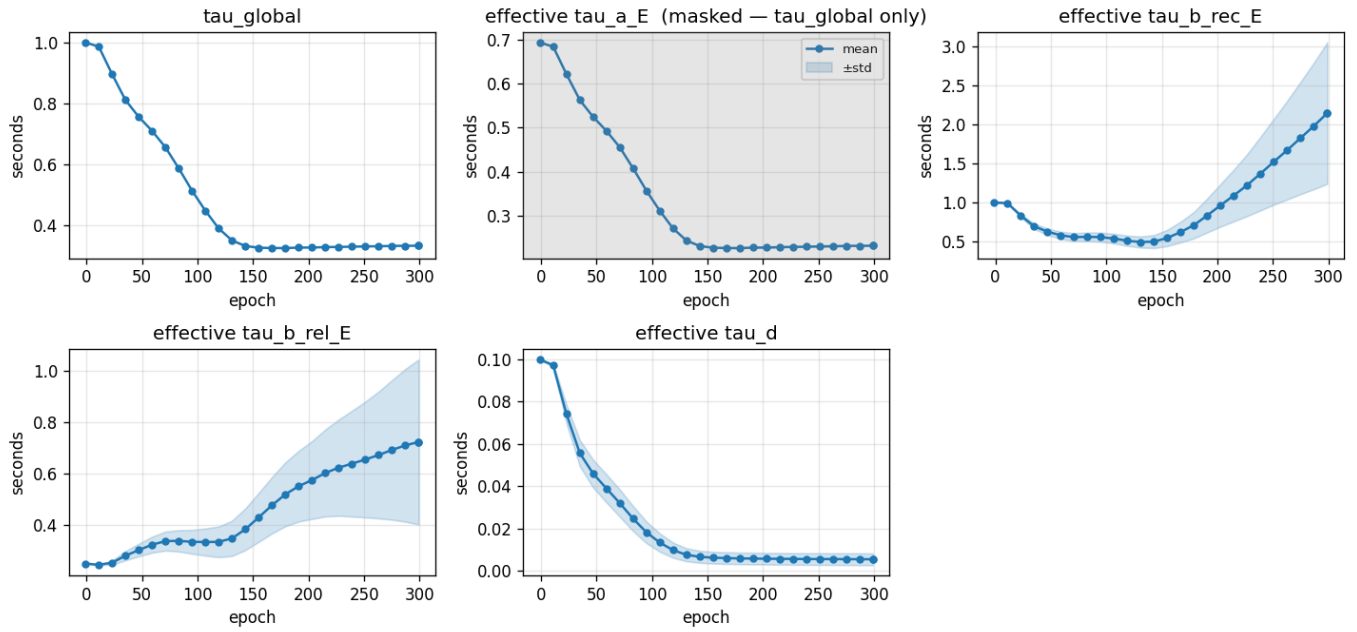


Figure 11: Tau evolution

rk4-frz-a0c0e-300e — recurrent E vs I weights (srnn-std-e-only-per-neuron)

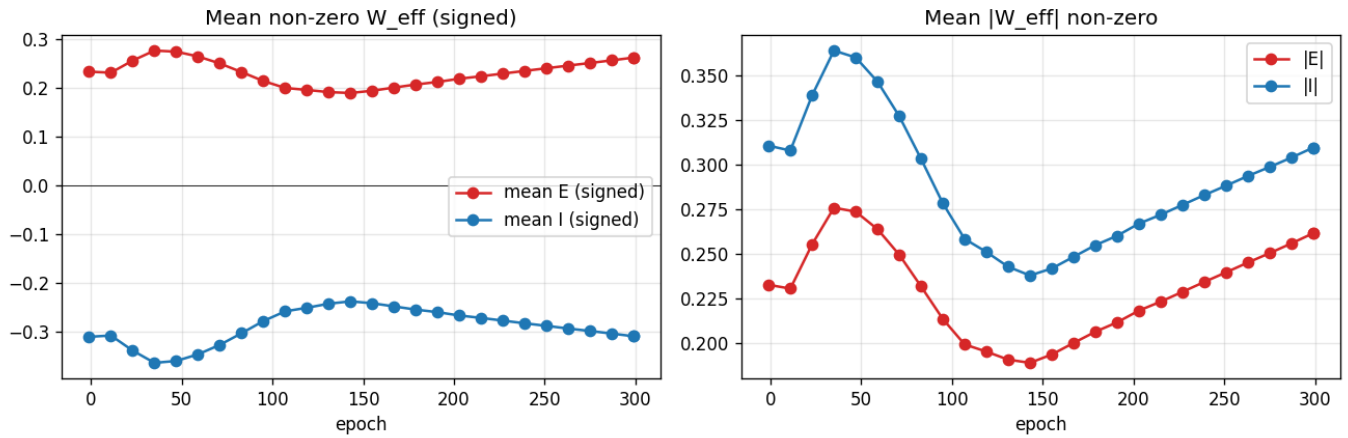


Figure 12: W_EI evolution

srnn-std-e-only-per-neuron — parameter table

```
# rk4-frz-a0c0e-300e - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.33416	-	-	scalar
tau_d (s)	+0.1	+0.0053372	0	0.002804	300
tau_b_rec_E (s)	+1	+2.1426	0	0.9039	150
tau_b_rel_E (s)	+0.25	+0.7232	0	0.321	150
W_eff (E src, signed)	+0.23267	+0.26155	0.07641	0.1251	15048
W_eff (I src, signed)	-0.31062	-0.30953	0.07724	0.1024	14977
W_eff (E src)	+0.23267	+0.26155	0.07641	0.1251	15048
W_eff (I src)	+0.31062	+0.30953	0.07724	0.1024	14977
W_in_eff (input neurons)	+0.00019944	-0.00017023	0.09994	0.03078	6675
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00074661	-0.0029953	0.06657	0.3847	6675
readout_bias	+0	-0.057122	0	0.2424	89
ic.ic (per-variant init)	+0.090198	+0.090198	0.3825	0.3825	900

srnn-std-e-only-skip-per-neuron

rk4-frz-a0c0e-300e — effective time constants (srnn-std-e-only-skip-per-neuron)

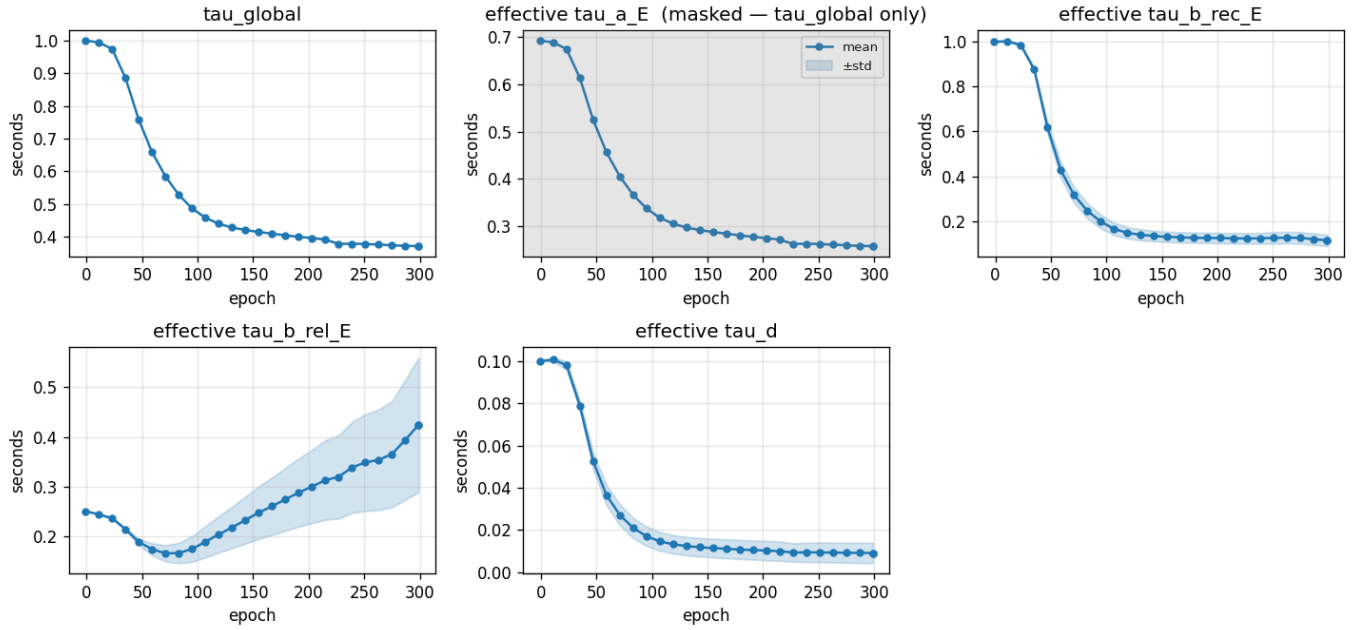
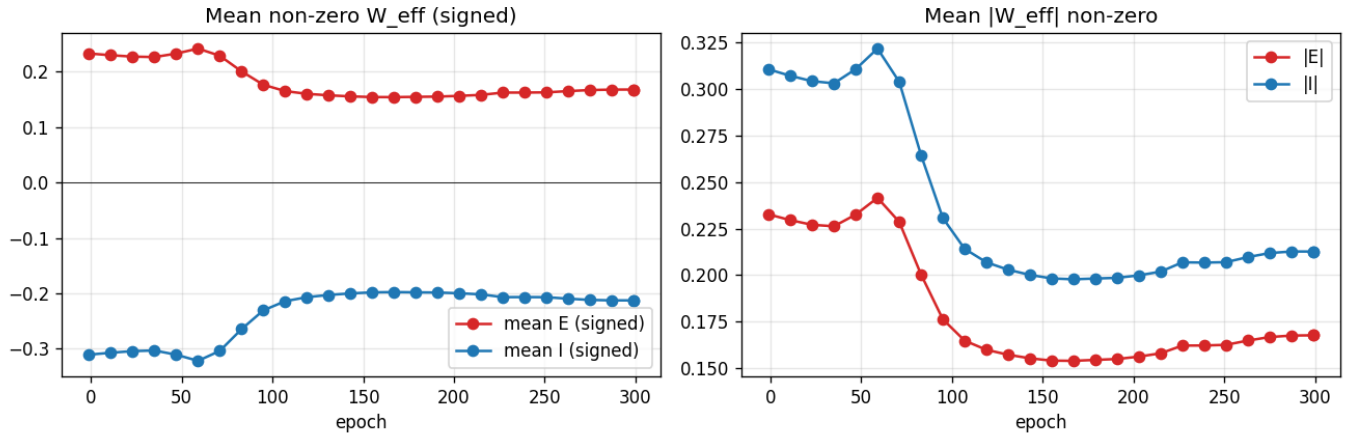


Figure 13: Tau evolution

rk4-frz-a0c0e-300e — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

Figure 14: W_{EI} evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# rk4-frz-a0c0e-300e - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.37143	-	-	scalar
tau_d (s)	+0.1	+0.0091822	0	0.004845	300
tau_b_rec_E (s)	+1	+0.11387	0	0.0255	150
tau_b_rel_E (s)	+0.25	+0.42433	0	0.1359	150
W_eff (E src, signed)	+0.23267	+0.16768	0.07641	0.06635	15048
W_eff (I src, signed)	-0.31062	-0.2127	0.07724	0.06666	14977
W_eff (E src)	+0.23267	+0.16768	0.07641	0.06635	15048
W_eff (I src)	+0.31062	+0.2127	0.07724	0.06666	14977
W_in_eff (input neurons)	+0.00036119	-4.0943e-05	0.09956	0.0164	6675
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00055372	-0.0045908	0.06572	0.1673	6675
readout_bias	+0	+0.039199	0	0.089	89
ic.ic (per-variant init)	+0.082727	+0.082727	0.3836	0.3836	900